MTH 232(A and B) Dr. James Ashe
Quiz February 11, 2013 **Name:**

Justify your answers and show your work. This quiz will be worth two homework grades.

Complete the following steps for $f(x) = 24 - 2x$, *interval* $[0, 6]$, *and* $n = 3$.

1. Sketch the graph of the function on the given interval and illustrate the *right Riemann sums* using n subintervals. Clearly label the gridpoints.

2. Calculate the *right Riemann sums* for the given interval and number of subintervals, n.

The figure shows the area of regions bounded by the graph of $f(x)$ *and the* $x-$*axis. Evaluate the following integrals.*

3. $\displaystyle\int_0^8 f(x)\,dx$

4. $\displaystyle\int_5^8 2f(x)\,dx$

5. $\displaystyle\int_0^5 f(x)\,dx - \int_8^0 f(x)\,dx$

6. Write the series in summation notation: $2 + 4 + 6 + 8 + 10$.

Use area to evaluate the definite integral.

7. $\int_{-9}^{9} \sqrt{81 - x^2}\, dx$

Use the value of the first integral I to evaluate the given integral. $I = \int_0^1 \left(x^3 - 5x\right)\, dx = -\frac{7}{4}$.

8. $\int_0^1 \left(10x - 2x^3\right)\, dx$

Evaluate the following.

9. Let a and b be constants $\frac{d}{dx} \int_a^x f(t)\, dt$

10. $\frac{d}{dx} \int_x^2 \sqrt{t^5}\, dt$

11. $\int_1^2 (x^2 - 8x + 1)\, dx$

12. $\displaystyle\int \frac{x-\sqrt{x}}{x^3}\,dx$

13. $f(x) = \cos 3x$

a. $\displaystyle\int \cos 3x\,dx$

b. $\displaystyle\int_{-\pi/3}^{\pi/3} \cos 3x\,dx$

14. Find the solution to the following intital value problem. $g'(x) = 4x\left(x^3 - \frac{1}{4}\right)$ and $g(1) = 2$.

15. A bowling ball rolls down a ramp at a speed of $v(t) = -3t^2$ feet per second (where t is seconds). The distance of the ball from the top of the ramp is 4 feet after 2 seconds. Find the function giving the distance of the ball from the top of the ramp at t seconds.